

Notes: Morris & Shin (AER, 2002)

Social Value of Public Information

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1 Big picture

- **Question.** Is more public information always socially valuable when agents have both a fundamentals motive and a coordination motive?
- **Core answer.** No. Public information improves knowledge of fundamentals, but it also becomes a common focal point for higher-order beliefs.
- **Main mechanism.** Agents rationally overweight public signals because they are useful for predicting what other agents will do.
- **Main result.** Private precision always raises welfare, but public precision can reduce welfare when the coordination motive is strong and private information is already precise.
- **Policy implication.** High-visibility disclosures by central banks and public agencies can do harm if noisy public signals coordinate behavior away from fundamentals.

2 Incremental contribution of the paper

- **Beyond single-agent information value.** The paper shows that the standard result that more information is valuable can fail in strategic settings.
- **Beyond pure sunspots.** Public information is not treated as extrinsic; it conveys fundamentals information but also coordinates beliefs.
- **Beyond irrational overreaction.** Agents are fully rational; the overreaction to public information is equilibrium behavior.
- **Unique-equilibrium welfare result.** The welfare loss does not rely on multiplicity or equilibrium selection.
- **Policy contribution.** The paper gives a rigorous reason why transparency policy must consider signal noise, private information, and strategic complementarity.

3 Model objects and measurement

The state is θ . Agent i chooses action a_i . The strategic-complementarity parameter is $r \in (0, 1)$. Public information is $y = \theta + \eta$ with precision $\alpha = 1/\sigma_\eta^2$. Private information is $x_i = \theta + \varepsilon_i$ with precision $\beta = 1/\sigma_\varepsilon^2$.

Interpretation: The paper has no data. The “measurement” objects are theoretical: precision, welfare, public-signal noise, private-signal noise, and the coordination weight.

4 Models and theoretical specifications

4.1 Equation Panel (1)–(2): payoff, social welfare, and best response

$$u_i(a, \theta) = -(1-r)(a_i - \theta)^2 - r(L_i - \bar{L}), \quad L_i = \int_0^1 (a_j - a_i)^2 dj.$$

$$W(a, \theta) = - \int_0^1 (a_i - \theta)^2 di, \quad a_i = (1-r)E_i(\theta) + rE_i(\bar{a}).$$

Interpretation: The social planner values closeness to fundamentals. The individual also values closeness to other actions. This creates the wedge.

4.2 Public and private information

If only public information is available,

$$y = \theta + \eta, \quad a_i(y) = y, \quad E[W \mid \theta] = -\sigma_\eta^2.$$

With public and private signals,

$$E_i(\theta) = \frac{\alpha y + \beta x_i}{\alpha + \beta}, \quad a_i = \frac{\alpha y + \beta(1-r)x_i}{\alpha + \beta(1-r)}.$$

Interpretation: The equilibrium action puts more weight on the public signal than the Bayesian posterior because public information is useful for coordination.

4.3 Welfare derivatives

Private precision always helps:

$$\frac{\partial E(W \mid \theta)}{\partial \beta} > 0.$$

Public precision can hurt:

$$\frac{\partial E(W \mid \theta)}{\partial \alpha} \geq 0 \quad \text{iff} \quad \frac{\beta}{\alpha} \leq \frac{1}{(2r-1)(1-r)}.$$

Interpretation: When $r > 1/2$ and private information is precise, extra public precision can lower welfare.

5 Theoretical design and interpretation

- The continuum of agents removes aggregate idiosyncratic private noise.
- The equilibrium is unique for all parameter values.
- The welfare loss is not a failure of rationality.
- The main limitation is that the payoff function is stylized and the model is not an empirical test.

6 Figures, equation panels, and full interpretation

6.1 Equation Panel (1)–(2): Payoff, Social Welfare, and Individual Best Response

Read: The panel defines the payoff, the beauty-contest loss, social welfare, and the first-order condition.

Interpretation: The key wedge is that the beauty-contest term is privately relevant but socially zero-sum.

Economic message: The model shows why individually rational coordination can create socially inefficient attention to public signals.

What not to over-read: The parameter r is not merely psychology; it is a reduced-form measure of strategic complementarity.

agents, indexed by the unit interval $[0, 1]$. Agent i chooses an action $a_i \in \mathbb{R}$, and we write $\mathbf{a}$ for the action profile over all agents. The payoff function for agent i is given by

$$(1) \quad u_i(\mathbf{a}, \theta) \\ \equiv -(1-r)(a_i - \theta)^2 - r(L_i - \bar{L})$$

where r is a constant, with $0 < r < 1$ and

$$L_i \equiv \int_0^1 (a_j - a_i)^2 dj$$

$$\bar{L} \equiv \int_0^1 L_j dj.$$

The loss function for individual i has two components. The first component is a standard quadratic loss in the distance between the underlying state θ and his action a_i . The second component is the “beauty contest” term. The loss L_i is increasing in the average distance between i ’s action and the action profile of the whole population. There is an externality in which an individual tries to second-guess the decisions of other individuals in the economy. The parameter r gives the weight on this second-guessing motive. The larger is r , the more severe is the externality. Moreover, this spillover effect is socially inefficient in that it is of a zero-sum nature. In the game of second-guessing, the winners gain at the expense of the losers. Social welfare, defined as the (normalized) average of individual utilities, is

$$W(\mathbf{a}, \theta) \equiv \frac{1}{1-r} \int_0^1 u_i(\mathbf{a}, \theta) di \\ = - \int_0^1 (a_i - \theta)^2 di$$

so that a social planner who cares only about

Equation Panel (1)–(2): payoff, social welfare, and best response.

6.2 Equation Panel (3)–(9): Public Information Benchmark and Private-Signal Setup

Read: Public-only information always improves welfare; private signals introduce the possibility of overweighting public information.

Interpretation: The benchmark rules out an anti-transparency reading: public information is good when it is the only information.

Economic message: Disclosure value depends on the existing private information environment.

What not to over-read: The paper does not say that public information is always harmful.

$$(2) \quad a_i = (1 - r)E_i(\theta) + rE_i(\bar{a})$$

where $\bar{a}$ is the average action in the population (i.e., $\bar{a} = \int_0^1 a_j dj$) and $E_i(\cdot)$ is the expectation operator for player i . Thus each agent puts positive weight on the expected state and the expected actions of others. Note, however, that if θ is common knowledge, the equilibrium entails $a_i = \theta$ for all i , so that social welfare is maximized at equilibrium. So, when there is perfect information, there is no conflict between individually rational actions and the socially optimal actions. We now examine the case where θ is not known with certainty.

A. Public Information Benchmark

Consider the case where agents face uncertainty concerning θ , but they have access to public information. The state θ is drawn from an (improper) uniform prior over the real line, but the agents observe a *public signal*

$$(3) \quad y = \theta + \eta$$

where η is normally distributed, independent of θ , with mean zero and variance σ_η^2 . The signal y is “public” in the sense that the actual realization of y is common knowledge to all agents. They choose their actions after observing the realization of y . The expected payoff of agent i at the time of decision is then given by the conditional expectation:

$$(4) \quad E(u_i|y)$$

where $E(\cdot|y)$ is the common expectation operator. Conditional on y , both agents believe that θ is distributed normally with mean y and variance σ_η^2 . Hence, the best reply of i is

$$(5) \quad a_i(y) = (1 - r)E(\theta|y) + r \int_0^1 E(a_j|y) dj$$

where $a_i(y)$ denotes the action taken by agent i as a function of y . Since $E(\theta|y) = y$ and since the strategies of both agents are measurable with respect to y , we have $E(a_j|y) = a_j(y)$, so that in the unique equilibrium,

$$(6) \quad a_i(y) = y$$

for all i ; expected welfare, conditional on θ , is

$$\begin{aligned} E(W|\theta) &= -E[(y - \theta)^2|\theta] \\ &= -\sigma_\eta^2. \end{aligned}$$

Thus, the smaller the noise in the public signal, the higher is social welfare. We will now contrast this with the general case in which agents have private information as well as public information.

B. Private and Public Information

Consider now the case where, in addition to the public signal y , agent i observes the realization of a *private signal*:

$$(7) \quad x_i = \theta + \varepsilon_i$$

where noise terms ε_i of the continuum population are normally distributed with zero mean and variance σ_ε^2 , independent of θ and η , so that $E(\varepsilon_i \varepsilon_j) = 0$ for $i \neq j$. The private signal of one agent is not observable by the others. This is the sense in which these signals are private.

As before, the agents’ decisions are made after observing the respective realizations of their private signals as well as the realization of the public signal. Denote by

$$(8) \quad a_i(I_i)$$

the decision by agent i as a function of his information set I_i . The information set I_i consists of the pair (y, x_i) that captures all the information available to i at the time of decision.³

Let us denote by α the precision of the public information, and denote by β the precision of the private information, where

$$(9) \quad \begin{aligned} \alpha &= \frac{1}{\sigma_\eta^2} \\ \beta &= \frac{1}{\sigma_\varepsilon^2}. \end{aligned}$$

³ The notation in (8) makes explicit that the *strategy* of agent i in the imperfect-information game is a function that maps the information I_i to the action a_i . For any given strategy, a_i is therefore a random variable that is measurable on the partition generated by I_i .

Equation Panel (3)–(9): public benchmark and private information structure.

6.3 Equation Panel (10)–(15): Linear Equilibrium and Higher-Order Beliefs

Read: The panels derive the linear equilibrium and show how higher-order expectations emphasize public information.

Interpretation: Public information dominates because it anchors beliefs about beliefs.

Economic message: Public signals are coordination devices, not merely fundamentals signals.

What not to over-read: The mechanism does not require multiple equilibria.

Then, based on both private and public information, agent i 's expected value of θ is

$$(10) \quad E_i(\theta) = \frac{\alpha y + \beta x_i}{\alpha + \beta}$$

where we have used the shorthand $E_i(\cdot)$ to denote the conditional expectation $E(\cdot | I_i)$.

C. Linear Equilibrium

We will now solve for the unique equilibrium. We do this in two steps. We first solve for a linear equilibrium in which actions are a linear function of signals. We will follow this with a demonstration that this linear equilibrium is the unique equilibrium. Thus, as the first step, suppose that the population as a whole is following a linear strategy of the form

$$(11) \quad a_i(I_i) = \kappa x_i + (1 - \kappa)y.$$

Then agent i 's conditional estimate of the average expected action across all agents is

$$\begin{aligned} E_i(\bar{a}) &= \kappa \left(\frac{\alpha y + \beta x_i}{\alpha + \beta} \right) + (1 - \kappa)y \\ &= \left(\frac{\kappa\beta}{\alpha + \beta} \right) x_i + \left(1 - \frac{\kappa\beta}{\alpha + \beta} \right) y. \end{aligned}$$

Thus agent i 's optimal action is

$$\begin{aligned} (12) \quad a_i(I_i) &= (1 - r)E_i(\theta) + rE_i(\bar{a}) \\ &= (1 - r) \left(\frac{\alpha y + \beta x_i}{\alpha + \beta} \right) \\ &\quad + r \left[\left(\frac{\kappa\beta}{\alpha + \beta} \right) x_i + \left(1 - \frac{\kappa\beta}{\alpha + \beta} \right) y \right] \\ &= \left(\frac{\beta(r\kappa + 1 - r)}{\alpha + \beta} \right) x_i \\ &\quad + \left(1 - \frac{\beta(r\kappa + 1 - r)}{\alpha + \beta} \right) y. \end{aligned}$$

Comparing coefficients in (11) and (12), we therefore have

$$\kappa = \frac{\beta(r\kappa + 1 - r)}{\alpha + \beta}$$

from which we can solve for κ :

$$\kappa = \frac{\beta(1 - r)}{\beta(1 - r) + \alpha}.$$

Thus, the equilibrium action a_i is given by

$$(13) \quad a_i(I_i) = \frac{\alpha y + \beta(1 - r)x_i}{\alpha + \beta(1 - r)}.$$

D. Uniqueness of Equilibrium

The argument presented above establishes the existence of a linear equilibrium. We will follow this by showing (through a “brute force” solution method) that the linear equilibrium we have identified is the unique equilibrium. In doing so, we establish the role of higher-order expectations in this model. In particular, we note that, if someone observes a public signal that is worse than her private signal, then her expectation of others' expectations of θ is lower than her expectation of θ (i.e., it is closer to the public signal than her own expectation). This in turn implies that if we look at n th order expectations about θ (i.e., someone's expectation of others' expectations of others' expectations of θ), then this approaches the public signal as n becomes large. Higher-order expectations depend only on public signals.

Recall that player i 's best response is to set

$$a_i = (1 - r)E_i(\theta) + rE_i(\bar{a}).$$

Substituting and writing $\bar{E}(\theta)$ for the average expectation of θ across agents we have

$$\begin{aligned} (14) \quad a_i &= (1 - r)E_i(\theta) + (1 - r)rE_i(\bar{E}(\theta)) \\ &\quad + (1 - r)r^2E_i(\bar{E}^2(\theta)) + \dots \\ &= (1 - r) \sum_{k=0}^{\infty} r^k E_i(\bar{E}^k(\theta)). \end{aligned}$$

In order to evaluate this expression, and check that the infinite sum is bounded, we must solve

explicitly for $E_i(\bar{E}^k(\theta))$. Recall that player i 's expected value of θ is

$$(15) \quad E_i(\theta) = \frac{\alpha y + \beta x_i}{\alpha + \beta}.$$

Thus, the average expectation of θ across agents is

$$\bar{E}(\theta) = \int_0^1 E_i(\theta) di = \frac{\alpha y + \beta \theta}{\alpha + \beta}.$$

Now player i 's expectation of the average expectation of θ across agents is

$$\begin{aligned} E_i(\bar{E}(\theta)) &= E_i \left(\frac{\alpha y + \beta \theta}{\alpha + \beta} \right) \\ &= \frac{\alpha y + \beta \left(\frac{\alpha y + \beta x_i}{\alpha + \beta} \right)}{\alpha + \beta} \\ &= \frac{((\alpha + \beta)^2 - \beta^2)y + \beta^2 x_i}{(\alpha + \beta)^2} \end{aligned}$$

and the average expectation of the average expectation of θ is

$$\begin{aligned} \bar{E}^2(\theta) &= \bar{E}(\bar{E}(\theta)) \\ &= \frac{((\alpha + \beta)^2 - \beta^2)y + \beta^2 \theta}{(\alpha + \beta)^2}. \end{aligned}$$

More generally, we have the following lemma.

LEMMA 1: For any k , $\bar{E}^k(\theta) = (1 - \mu^k)y + \mu^k \theta$ and $E_i(\bar{E}^k(\theta)) = (1 - \mu^{k+1})y + \mu^{k+1}x_i$ where $\mu = \beta/(\alpha + \beta)$.

PROOF:

The proof is by induction on k . We know from (15) that the lemma holds for $k = 1$. Suppose that it holds for $k - 1$. Then,

and

$$\begin{aligned} E_i(\bar{E}^k(\theta)) &= (1 - \mu^k)y + \mu^k \left(\frac{\alpha y + \beta x_i}{\alpha + \beta} \right) \\ &= (1 - \mu^{k+1})y + \mu^{k+1}x_i \end{aligned}$$

which proves Lemma 1. Now substituting the expression from Lemma 1 into equation (14), we obtain

$$\begin{aligned} a_i &= (1 - r) \sum_{k=0}^{\infty} r^k [(1 - \mu^{k+1})y + \mu^{k+1}x_i] \\ &= \left(1 - \frac{\mu(1 - r)}{1 - r\mu} \right) y + \left(\frac{\mu(1 - r)}{1 - r\mu} \right) x_i \\ &= \frac{\alpha y + \beta(1 - r)x_i}{\alpha + \beta(1 - r)}. \end{aligned}$$

This is exactly the unique linear equilibrium we identified earlier.

E. Lucas-Phelps Island Economy Model

We conclude this section by drawing out the parallels between the equilibrium in our model and features of the celebrated Lucas-Phelps “island economy” model.⁴ To do this, let each index i refer to an island whose supply y_i^s of the single consumption good is given by

$$y_i^s = b[a_i - E_i(\bar{a})]$$

where a_i is the price on island i , $\bar{a}$ is the average price across all islands, and $b > 0$ is a supply parameter. The demand y_i^d on island i is given by

$$y_i^d = c[E_i(\theta) - a_i]$$

where θ is the money supply and $c > 0$ is the slope parameter for demand. Market-clearing then implies

$$a_i = (1 - r)E_i(\theta) + rE_i(\bar{a})$$

Lemma 1: higher-order expectations.

6.4 Equation Panel (16)–(20) and Figure 1: Welfare Derivatives and Social Welfare Contours

Read: Private precision always improves welfare; public precision can reduce welfare. Figure 1 shows the region where increasing public precision is harmful.

Interpretation: The welfare loss comes from common noisy mistakes receiving too much weight.

Economic message: Transparency can be harmful when public disclosures are noisy and agents already have precise private information.

What not to over-read: The result is not a case for deliberate inaccuracy; it is a warning about noisy high-salience disclosures.

increased precision of private and public information.

Welfare is always increasing in the precision of the private signals. We can see this by differentiating (17) with respect to β , the precision of the private signals. We have

$$(18) \quad \frac{\partial E(W|\theta)}{\partial \beta} = \frac{(1-r)[(1+r)\alpha + (1-r)^2\beta]}{[\alpha + \beta(1-r)]^3} >$$

Thus, increased precision of private information enhances welfare unambiguously. The same cannot be said of the effect of increased precision of the public signal. The derivative of (17) with respect to α is

$$(19) \quad \frac{\partial E(W|\theta)}{\partial \alpha} = \frac{\alpha - (2r-1)(1-r)\beta}{[\alpha + \beta(1-r)]^3}$$

so that

$$(20) \quad \frac{\partial E(W|\theta)}{\partial \alpha} \geq 0$$

if and only if

$$\frac{\beta}{\alpha} \leq \frac{1}{(2r-1)(1-r)}.$$

When $r > 0.5$, there are ranges of the parameters

Equation Panel (16)–(20): welfare derivatives.

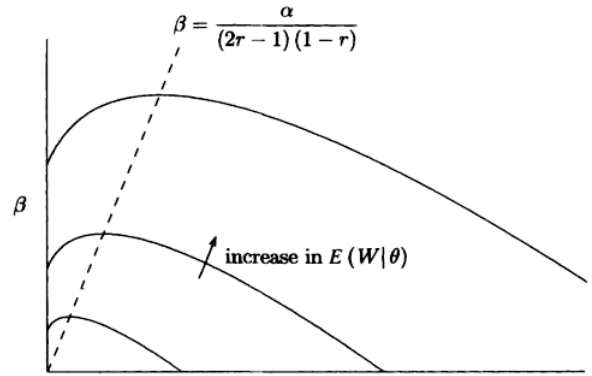


Figure 1: Social welfare contours.

6.5 Equation Panel (21): Decomposition of Action and Public-Signal Overreaction

Read: The equilibrium action equals the Bayesian posterior plus an overreaction term toward the public signal.

Interpretation: The extra term is the cleanest expression of the coordination distortion.

Economic message: The public signal is privately useful for forecasting others but socially costly when it coordinates mistakes.

What not to over-read: Overreaction is relative to the social benchmark, not relative to individual optimality.

in (α, β) -space. The curves are the set of points that satisfy $E(W|\theta) = C$, for constants C . As can be seen from Figure 1, when $\beta > \alpha/[(2r-1)(1-r)]$, the social welfare contours are upward-sloping, indicating that welfare is *decreasing* in the precision of public information.

What is the intuition for this result? Observe that equation (13) can be rewritten as

$$(21) \quad a_i = \frac{\alpha y + \beta(1-r)x_i}{\alpha + \beta(1-r)} \\ = \frac{\alpha y + \beta x_i}{\alpha + \beta} + (y - x_i) \\ \times \left(\frac{\alpha}{\alpha + \beta} \right) \frac{\beta r}{\alpha + \beta(1-r)}.$$

Equation Panel (21): action decomposition.

6.6 Equation Panel (23)–(24): Strategic Substitutes Extension

Read: With strategic substitutes, agents overreact to private information rather than public information.

Interpretation: The sign of strategic interaction determines which type of information is overused.

Economic message: Information policy is environment-specific.

What not to over-read: The main result applies most directly to strategic-complementarity environments.

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equilibrium can be obtained from the method of comparing coefficients to yield

$$(23) \quad a_i = \frac{1}{1+\rho} \left(\frac{\beta(1+\rho)}{\alpha + \beta(1+\rho)} x_i + \frac{\alpha}{\alpha + \beta(1+\rho)} y \right).$$

The symmetric-information benchmark solution is when $a_i = a_j$ for all i, j which gives $a_i = E_i(a)$, so that

$$(24) \quad a_i = \frac{1}{1+\rho} \left(\frac{\beta x_i + \alpha y}{\beta + \alpha} \right).$$

Comparing (23) and (24), the introduction of strategic substitutability implies that agents now overreact to *private information* x_i relative to the symmetric-information benchmark. Players accentuate their differences in order to avoid playing similar actions to other players.

An early paper by Roy Radner (1962) gives a nice instance of this.⁷ He examines the problem where two members of a team aim to minimize the loss function

$$(a_1 - \theta)^2 + (a_2 - \theta)^2 + 2q(a_1 - \theta)(a_2 - \theta).$$

In other words, the loss is increasing in the product of the two errors. This gives rise to strategic substitutability between the two team members, so that the optimal decisions put less weight on the public information and more weight on the private information as compared to the individual decision. The choice of output in a Cournot model examined by Robert M. Townsend (1978) also

deviation of the *aggregate* level of activity from the true state θ . Consider a finite player version of our framework, where the principal's objective is to minimize

$$\left(\frac{1}{n} \sum_i a_i - \theta \right)^2$$

so that the objective for the principal is to set the *average* action as close as possible to θ . Suppose that all agents follow a linear strategy and set their action according to

$$a_i = \kappa x_i + (1 - \kappa)y$$

where y is the public signal, and x_i is i 's private signal. Then the expected loss for the principal at θ is

$$E \left[\left(\frac{\kappa}{n} \sum_{j=1}^n x_j + (1 - \kappa)y - \theta \right)^2 \right] \\ = E \left[\left(\frac{\kappa}{n} \sum_{j=1}^n e_j + (1 - \kappa)\eta \right)^2 \right] \\ = \frac{\kappa^2}{n\beta} + \frac{(1 - \kappa)^2}{\alpha}.$$

The value of κ that minimizes the principal's loss is

$$\kappa = \frac{n\beta}{n\beta + \alpha}$$

Equation Panel (23)–(24): strategic substitutes extension.

6.7 Concluding Policy Panel: Disclosure as a Double-Edged Instrument

Read: The conclusion applies the model to central bank communication and high-visibility public disclosures.

Interpretation: Public information can coordinate beliefs beneficially or harmfully depending on its noise and on private-sector information.

Economic message: Transparency is a design problem involving timing, precision, and salience.

What not to over-read: The policy implication is not secrecy; it is careful communication design.

increased vulnerability mentioned by Krugman is an entirely rational response by individual actors, but is socially inefficient.

The challenge for central banks and other public organizations is to strike the right balance between providing timely and frequent information to the private sector so as to allow it to pursue its goals, but to recognize the inherent limitations in any disclosure and to guard against the potential damage done by noise. This is a difficult balancing act at the best of times, but this task is likely to become even harder. As central banks' activities impinge more and more on the actions of market participants, the latter have reciprocated by stepping up their surveillance of central banks' activities and pronouncements. The intense spotlight trained on the fledgling European Central Bank and the ECB's delicate relationship with the press and private sector market participants illustrate well the difficulties faced by policy makers.

In the highly sensitized world of today's financial markets populated with Fed watchers, economic analysts, and other commentators of the economic scene, disclosure policy assumes great importance. Our results suggest that private sources of information may actually *crowd out* the public information by rendering the public information detrimental to the policy maker's goals. The heightened sensitivities of the market could magnify any noise in the public information to such a large extent that public information ends up by causing more

Concluding policy panel.

7 Short summary

- The paper studies public information in a beauty-contest economy.
- Public information can be socially harmful because agents coordinate too strongly on it.
- Private information always improves welfare.
- Public information improves welfare only when its fundamentals value dominates the coordination distortion.
- The mechanism is fully rational and works through higher-order beliefs.

8 Expanded conclusion

8.1 Core conclusion

Public information is a double-edged instrument. It teaches agents about fundamentals and coordinates their actions. The second role can create welfare losses when a noisy public signal becomes excessively salient.

8.2 Incremental contribution revisited

The paper gives a precise welfare theorem for a problem that had often been discussed informally: public announcements can move markets too much. It formalizes this without irrationality or multiple equilibria.

8.3 Policy implications

Central banks, statistical agencies, and regulators should ask not only whether information is accurate, but also whether it will become a focal point that crowds out dispersed private information.

8.4 Final takeaway

The right benchmark is not maximum transparency. It is optimal communication: disclose information in a way that balances fundamentals learning against the coordination distortion created by common noisy signals.